

21/09/26

Excel

1) What is Excel?

Microsoft Excel is a spreadsheet application developed by Microsoft that is used to store, organize, calculate, analyze and visualize data using rows, columns, formulas, functions, tables and charts.

2) Purpose of Excel?

- Clean Raw data
- Build Interactive dashboard
- Visualize data trends
- Manage Structured Information.

3) Types of data in Excel?

• Text: Contains words, names or characters.

Ex: Employee.

• Number: Contains numerical values used for calculation

Ex: 100

• Date: Represents a specific date.

Ex: 26/09/2026

• Time: Represents a specific time.

Ex: 6:30pm

• Logical: Represents a logical condition

Ex: TRUE, FALSE

• Formula: An expression used to perform calculation

Ex: A1+B1

- Error: Appears when excel encounters an error in a calculation or formula.
Ex: #DIV/0!, #VALUE!

Table

4. Excel's feature for data analysis

- ⇒ Data Extraction
- * Data Cleaning
 - Remove duplicate
 - handling the missing values
 - Removes the irrelevant data and adding necessary data.
 - Text columns find and replace.

5) Formulas and functions

* Formulas

- A formula is an expression created by user to perform calculation it begins with =

- Addition $\rightarrow = A1 + B1$
- Subtraction $\rightarrow = A1 - B1$
- Multiplication $\rightarrow = A1 * B1$
- Division $\rightarrow = A1 / B1$
- Performs multiple calculation $= (A1 + B1) * 2$

* Functions

- A function is a predefined formula in excel that performs a specific calculation.

- Sum
- Average
- Max
- Min
- Count
- Counta
- If
- Basic : Sum, Product, Round, Roundup, Rounddown, Abs, mod, INT, CEILING, FLOOR
- Statistical : AVERAGE, MEDIAN, MODE, MIN, MAX, COUNT, COUNTA, COUNTBLANK, COUNTIF, COUNTIFS
- Logical : IF, IFS, AND, OR, NOT, TRUE, FALSE, SWITCH
- Lookup : VLOOKUP, HLOOKUP, XLOOKUP, LOOKUP, INDEX, MATCH, XMATCH
- Text : Concat, Upper, Lower

② Data visualization

Data visualization is the process of representing data in graphical or visual form

- Charts : Bar, column, pie etc
- Conditional formatting
- Sparklines

- * Purpose.
 - Makes complex data easy to understand.
 - Helps identify trends & patterns
 - Makes comparisons easier
 - Helps identify outliers
 - Supports better decision making

- makes reports and dashboards more informative.

→ Pivot table

A pivot table is an excel tool used to summarise, analyse and organize large amount of data quickly. It allows you to rearrange data to see it from different perspective without changing the original data.

2) Sorting and filtering

* Sorting

Sorting is the process of arranging data in a specific order, such as ascending or descending order.

* Filtering

Filtering is a process of displaying only the data that meets specific conditions while temporarily hiding the other records.

3. Data validation

- Prevents incorrect data entry
- Maintain data accuracy and consistency
- Greater easy-to-use drop-down lists
- Reduces spelling and formatting errors
- Makes data analysis more reliable.

- 9) Advantages
- * Beginner friendly
 - * Fast for small to medium dataset
 - * widely used in companies
 - * Integrates with tools like Power BI
- 10) Disadvantages
- * Not suitable for large datasets
 - * Limited automation compared to coding to
 - * Manual errors are possible.